



Genome-wide identification of multifunctional laccase gene family in *Eucalyptus grandis*: potential targets for lignin engineering and stress tolerance

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Abstract

Key message A survey of the *Eucalyptus grandis* genome revealed the presence of 54 laccase genes. Their tissue-specific and stress-induced expression patterns suggest a role in lignification and adaptation to abiotic stresses.

Abstract Laccases are multicopper oxidases that play important roles in the oxidation of monolignols during lignin biosynthesis and are reported to be functionally involved in plant development and stress responses. In this study, a genome-wide survey of the *Eucalyptus grandis* genome revealed the presence of 54 putative *LAC* genes (referred as *EgrLAC*), which were assigned to six different phylogenetic groups. Among them, 17 were predicted to be potential targets of miR397, a negative regulator of lignin biosynthesis. Based on different RNA-Seq datasets, distinct organ/tissue expression patterns of the identified *EgrLAC* genes were ascertained. The vast majority, however, showed enriched expression in the vascular tissues of roots and stems. Additional expression profiling of selected *EgrLAC* genes revealed differential expression in response to oxidative and osmotic stresses, suggesting a role in abiotic stress responses. Parallel promoter analysis of *EgrLAC4*, a close homologue to the Arabidopsis lignin biosynthesis-related gene *AtLAC17*, revealed a vascular expression pattern, mostly associated with the phloem. Overall, our data point to an involvement of the identified *EgrLAC* genes in lignification and in *Eucalyptus* adaptation to abiotic stresses.

Keywords Laccase · Gene structure · Gene expression · Lignification · Abiotic stress · *Eucalyptus*

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Introduction

Laccases (EC 1.10.3.2) are multicopper oxidases that catalyze the one-electron oxidation of multiple substrates with the concomitant reduction of molecular oxygen to water (Mot and Silaghi-Dumitrescu 2012). These glycosylated proteins possess three active sites (namely Type 1, Type 2, and/or Type 3) that comprise four copper ions and catalyze the oxidation of a wide range of phenolic compounds.

In plants, these enzymes are encoded by a multigene family and, despite being implicated in different biological processes, a major role in lignin biosynthesis has been reported (reviewed in Wang et al. 2015a). The first evidence came from early studies showing that some plant laccases are expressed in lignifying tissues and able to catalyze the oxidative polymerization of lignin precursors (Sterjiades et al. 1992; Bao et al. 1993; Ranocha et al. 2002). This hypothesis was further corroborated by subsequent studies reporting the genetic manipulation of laccase genes in plants. Although

some discrepancies exist, in general, the disruption of specific laccase genes is frequently associated with reduced or abolished lignin accumulation (Berthet et al. 2011; Zhao et al. 2013; Wang et al. 2015b), while their transgenic overexpression often results in increased lignin content (Wang et al. 2008). Consistent with these evidences, reduced levels of lignin were observed in *Arabidopsis thaliana* and *Populus trichocarpa* overexpressing miR397, a known negative regulator of laccase expression (Lu et al. 2013; Wang et al. 2014). Overall, these results indicate that laccase genes linked to lignin biosynthesis represent potential targets for genetic manipulation of lignin content (Wang et al. 2015a).

Plant laccases have been also experimentally linked to environmental stress responses. First, a number of studies have demonstrated that laccase gene expression is often modulated in response to abiotic and biotic factors (Liang et al. 2006b; Liu et al. 2017; Wang et al. 2017). In rice, for example, the expression of several laccase genes is induced by abiotic stresses such as drought, salt and heavy metals (Liu et al. 2017). As a proof of concept, cotton plants with improved defense response to both pathogens and pests have been obtained by either overexpressing or silencing a cotton laccase (Hu et al. 2018).

Laccases are encoded by medium to large multigene families in herbaceous and woody plant species, being prominent in the genomes of *P. trichocarpa* (49 genes; Lu et al. 2013), *Linum usitatissimum* (45 genes; Le Roy et al. 2017) and *Gossypium* spp. (84 genes in *Gossypium hirsutum*; Balasubramanian et al. 2016). In general, laccase family members display differential expression in different vegetative and reproductive organs/tissues and, as already mentioned, in response to a range of environmental constraints (Cai et al. 2006; Liu et al. 2017; Wang et al. 2017). Intriguingly, a significant large number of laccase genes (88 in total) was preliminary annotated in the reference genome of *Eucalyptus grandis*, an economically important wood tree species, suggesting that this family underwent a substantial expansion in *Eucalyptus* compared to other rosid species such as *P. trichocarpa* and *A. thaliana* (Myburg et al. 2014). Highlighting the relevance of laccases in lignin biosynthesis in *Eucalyptus*, it was postulated, based on an expressed sequence tag (EST) database survey, that laccases play a more prevalent role in lignification than peroxidases (Harakava 2005). Despite requiring additional confirmation, the apparent complexity of the laccase gene family in *Eucalyptus* suggests these enzymes can play functional roles in a variety of processes associated or not with lignification, and therefore, merits further investigation.

Eucalyptus is a fast-growing tree with great economic importance for the pulp and paper industry. Apart from its traditional use for paper pulp manufacturing, *Eucalyptus* has gained recent attention as a source of woody biomass for the production of biofuels (Rockwood et al. 2008). In

both cases, lignin has a highly negative impact in the corresponding industrial processes and its removal is costly and ecologically unfriendly. Therefore, the discovery of novel targets for lignin engineering in this important forest tree is of pivotal interest.

In this study, a genomic survey of the laccase genes present in the *E. grandis* genome was performed, and upon correction of previous annotation inconsistencies, a novel overview of this gene family is provided. To gain further insights into the functional/structural roles of the identified *Eucalyptus* laccases, we combined in-house and publicly available RNA-Seq data with RT-qPCR analyses to determine their organ/tissue-specific expression patterns. In addition, to corroborate a possible involvement in abiotic stress tolerance, the stress-responsive expression profiles of a set of laccase genes were evaluated. The obtained results represent an interesting start point to elucidate the roles of laccase genes in *Eucalyptus* and a source to identify laccases potentially involved in lignification for future application in forest tree biotechnology.

Materials and methods

Plant material and treatments

A commercial clone of *E. grandis* (SP0791), kindly provided by Suzano Papel e Celulose S.A., Brazil, was used in all assays. Plantlets were maintained in a grow chamber under controlled conditions (25° C and 14 h light/10 h dark). Tissue-specific expression of the identified genes was validated by RT-qPCR using freshly harvested vegetative organs/tissues (leaf, stem and root) obtained from independent 4-month-old plantlets ($n = 5$) grown in tubes under the described conditions. To reduce plant to plant variation, each harvested group of organ/tissue samples was pooled for total RNA extraction. Organ/tissue sampling to obtain the high-quality RNA used for RNA-Seq library preparation was performed in the same way, but in this case, total RNA was extracted from each sample individually ($n = 5$) and then pooled in equimolar amounts into a bulk representing each organ/tissue.

To investigate the oxidative and osmotic stress-inducible expression patterns of a set of selected *Eucalyptus* laccases genes, stress imposition was performed essentially as described (Rodrigues et al. 2016). Briefly, 4-month-old plantlets ($n = 15$) grown in Hoagland's nutrient solution 75% were treated with 100 mM H₂O₂ (for oxidative stress induction) or PEG8000 (for osmotic stress induction; osmotic potential of −0.8 MPa) for 6, 12 and 24 h. Plantlets ($n = 15$) maintained in Hoagland's solution 75% were used as control. Stem and root samples from five randomized treated and control plantlets were collected at the indicated time

points for each treatment, and independently pooled before RNA extraction. In all steps, collected fresh samples were immediately frozen in liquid nitrogen and stored at -80°C until total RNA extraction.

Identification of the laccase gene family from *E. grandis*

In the first step, the members of the *E. grandis* laccase gene family were identified by downloading all the sequences annotated as laccases in the *Eucalyptus* reference genome v2.0 (Myburg et al. 2014) available at Phytozome (<https://www.phytozome.net/>). To remove redundant sequences, the deduced amino acid sequences of the putative proteins were aligned by ClustalW using the MEGA Software package with default parameters. Subsequently, the identified sequences were individually inspected for the presence of laccase (IPR017761) and conserved multicopper oxidase types 1–3 (IPR001117, IPR011706, IPR011707) domains using the plug-in InterProScan available at Geneious 10.1.3 (<https://www.geneious.com>; Kearse et al. 2012) with default parameters. The presence of such domains was further validated using the Pfam conserved domain database (<https://pfam.xfam.org/>). The resulting sequences were reanalyzed in the National Center for Biotechnology Information (NCBI; <https://www.ncbi.nlm.nih.gov>) and Phytozome using BLASTp; the retrieved sequences were designated *EgrLAC* and numbered according to their gene IDs.

In silico characterization of the identified laccases

For exon/intron structure determination, the predicted coding sequence (CDS) of each gene was compared with its corresponding genomic sequence using the Gene Structure Display Server (GSDS 2.0; <https://gsds.cbi.pku.edu.cn/>). The chromosomal location was determined using the genomic annotation available at Phytozome. Tandem and segmental duplication events were retrieved using MCScan available at the Plant Genome Duplication Database (PGDD; <https://chibba.agtec.uga.edu/duplication/mcscan/>) with the default parameters. The physicochemical properties (protein length, molecular mass and isoelectric point) of the predicted proteins were determined employing the tools available on the ExPasy web server (<https://www.expasy.org/tools/>) and the online software Protein Isoelectric Point Calculator (<https://isoelectric.org/>). MicroRNA-target prediction was performed using the online target prediction tool psRNATarget (<https://plantgrn.noble.org/psRNATarget/>).

Phylogenetic analyses

A phylogenetic tree was first constructed using neighbor-joining method implemented in MEGA version 5.05 using

the full-length amino acid sequences of the 54 identified *EgrLAC*s. A bootstrap analysis of 1000 replicates was used. Subsequently, the phylogenetic relationships among the laccases proteins from *E. grandis*, *P. thricocharpa* and *A. thaliana* were determined. For that, the corresponding amino acid sequences were aligned using ClustalΩ (Sievers et al. 2011) implemented in SeaView (Gouy et al. 2010). Retained alignment blocks were obtained by GBLOCKS (Castresana 2000) using the following options: “Allow smaller final blocks” and “Allow gap positions within the final blocks”. Model testing was done using Prottest3 (Darriba et al. 2011). For that, all available models (120) were tested with four categories of rate variation and the best model was selected via the Akaike information criterion (AIC). The best model was configured in PhyML through SeaView user interface to obtain the maximum likelihood tree. The accuracy of branching was obtained by the approximate likelihood ratio test (LRT) method. The nodes supported by aLRT $< 50\%$ were hidden.

EgrLAC expression profiling using RNA-Seq data

The *EgrLAC* organ/tissue expression patterns were first determined using publicly available RNA-Seq data (www.phytozome.net) generated from six different vegetative organs/tissues (young and mature leaves, shooting tips, phloem, immature xylem and xylem) collected from a hybrid clone of *E. grandis* $\times$ *E. urophylla* (Mizrachi et al. 2010). Transcript abundance was expressed as unit of normalized FPKM (fragments per kilobase of exon per million reads mapped), and $\log_2$ (FPKM + 1) values were used for heat map generation.

Concurrently, the expression of the corresponding genes was also investigated using in-house RNA-Seq datasets generated from root, stem and leaf samples collected from 4-month-old *E. grandis* plantlets (Fuchs-Ferraz, unpublished; Sequence Read Archive accessions SRR1481484; SRR1481485; SRR1481487; SRR1481488; SRR1481490 and SRR1481491). Briefly, the sequencing libraries were prepared using the RNA bulk of different organ/tissue samples (leaf, stem and root) and the TruSeq RNA sample Prep Kit (Illumina) according to manufacturer’s instructions. After quantification and quality verification, libraries were sequenced on the Illumina MiSeq platform using a paired-end 300 base pair (bp) run. For the expression profiling, the RPKM (reads per kilobase of exon per million reads mapped) values of each target gene in a respective library were $\log_2$ transformed ($\log_2\text{RPKM} + 1$) and used to infer gene expression in the corresponding organ/tissue. The heat maps were constructed using the MeV Software (Multiple Experiment Viewer 4.9). Complementary expression analyses were further performed using The *Eucalyptus* Genome Integrative Explorer (EucGenIE;

<https://eucgenie.org/>). In addition, seven *EgrLAC* genes (Table S1) were randomly selected for validation of their organ/tissue expression using RT-qPCR as described above. For that, total RNA extracted from root, stem and leaf samples harvested from independent 4-month-old *E. grandis* plantlets ($n = 5$; pooled for analysis) grown under control conditions were employed.

Selection of stress-responsive genes for expression analysis

In silico analyses were carried out in a private *Eucalyptus* EST database (Scylla EST; Suzano Papel e Celulose) aiming to identify laccase genes potentially responsive to abiotic stresses. This database is composed of nonredundant EST sequences from cDNA libraries constructed using mRNA extracted from different organs/tissues of *Eucalyptus* grown under normal or stressed conditions. The abiotic stress-related libraries, including cold and drought stresses, were searched for the occurrence of ESTs corresponding to laccases, and those highly represented were considered stress-responsive. Among them, five genes (*EgrLAC4*, *EgrLAC14*, *EgrLAC35*, *EgrLAC45* and *EgrLAC52*) were randomly selected to experimentally verify their stress-responsive expression. Gene-specific primers were designed (Table S1) and inspected using PerlPrimer (<https://perlprimer.sourceforge.net/>). Additional BLAST analyses in the *Eucalyptus* reference genome were performed to confirm the primer specificity.

Total RNA extraction and quantitative RT-qPCR

Total RNA was extracted using the CTAB method with minor modifications (Korimbocus et al. 2002). The extracted RNA (1 µg) was treated with RNase free DNase I (MBI Fermentas) and employed for first-strand cDNA synthesis using the High Capacity[®] reverse transcriptase kit (Life Technologies) and random primers according to manufacturer's instructions. All qPCRs were carried out in a final volume of 10 µl using SYBR Green Master Mix (Applied Biosystems) and the following cycling conditions: initial denaturation at 95 °C for 5 min, followed by 45 cycles of 95 °C for 15 s and 60 °C for 1 min. Each reaction was performed in triplicate and a gene coding for a putative transcription elongation factor SII (*EgrTEF*; de Oliveira et al. 2012) was employed as endogenous control for normalization. Relative expression was determined using the $2^{-\Delta\Delta C_t}$ method as described (Livak and Schmittgen 2001). Statistical analyses were performed with GraphPad Prism software (www.graphpad.com) and differences with p values < 0.05 were considered statistically significant.

Promoter cloning and histochemical analysis

The 5'-flanking region of *EgrLAC4* was PCR-amplified using genomic DNA extracted from leaves of a commercial clone of *E. grandis* (SP0791) and two gene-specific primers (*EgrLAC4*-F—5'-TTGAACATAGGCATTATTTC-3' and *EgrLAC4*-R—5'-CGGCTGTTCTTGCCCC-3') designed based on the genomic sequence of *EgrLAC4* available in Phytozome. The resulting amplification product (1.38 kb) was gel purified, cloned into pGEM-Teasy (Promega) and sequenced.

For the construction of the expression cassette, the amplified fragment inserted into pGEM-Teasy was re-amplified using *Pfu* DNA polymerase and the primers *EgrLAC*-BamHI (5'-CGCGGATCCTTGAACATAGGCATTAT-3') and *EgrLAC*-NcoI (5'-CGCCATGGCGGCTGTTCTTGCC-3'), which contain *Bam*HI and *Nco*I recognition sites (underlined), respectively. The PCR-derived product was then digested by the mentioned enzymes, gel purified and cloned upstream of the promoterless *uidA* reporter gene (*GUS*) of the binary vector pCambia-1381z (Cambia).

The resulting plasmid was introduced into *Agrobacterium tumefaciens* strain GV3101, which was then used to transform *A. thaliana* (Col-0) employing the floral dip method (Zhang et al. 2006). Putative transformants were selected in agar plates containing MS (Murashige and Skoog 1962) medium supplemented with hygromycin (50 µg/ml). T-DNA integration in the regenerated lines was confirmed by PCR using genomic DNA extracted from leaves and primers complementary to the amplified promoter (*EgrLAC*-BamHI) and to the *GUS* coding region (5'-AGAACACACGAAATAAAGTAATCAG-3'), respectively. Only self-pollinated T2 progenies were employed in subsequent assays. Wild-type *A. thaliana* (Col-0) and a transgenic line harboring a *CaMV* 35S promoter::*GUS* fusion (*p35S::GUS*) were used as negative and positive controls, respectively. Promoter activity was assessed by histochemical *GUS* staining essentially as described (Jefferson et al. 1987). After staining, chlorophyll was bleached by a series of 70% ethanol washes and then photographed.

Results

As a starting point for our genome-wide analysis, the 88 sequences putatively annotated as laccases in the *E. grandis* genome (Myburg et al. 2014) were employed. After the removal of the redundant sequences, this dataset was reduced to 80 sequences. Subsequently, the encoded proteins of the remaining sequences were inspected for the presence of the conserved multicopper oxidase types 1–3 (type 1, IPR001117; type 2, IPR011706; type 3, IPR011707) and laccase (IPR017761) domains typically present in well-studied

laccases. As a result, only 54 of them were retained and indeed predicted to encode typical laccases in *Eucalyptus*. These 54 laccase genes were numbered according to their gene IDs and designated *EgrLAC1* through *EgrLAC54* (Table 1).

The identified *EgrLAC* genes are found dispersed on 9 (1, 2, 3, 5, 6, 7, 8, 10 and 11) of the 11 chromosomes of *E. grandis* (Table 1). Of note, 11 genes are spread out along chromosome 5, while 8 genes are located on chromosome 1. On the other hand, a single gene (*EgrLAC36*) is located on chromosome 8. Besides the *EgrLAC* genes mapped on chromosomes, five of them are associated with scaffolds (Table 1).

A phylogenetic tree constructed using neighbor joining revealed that the 54 *EgrLACs* could be clustered into different subgroups according to the similarity of their deduced amino acid sequences (Fig. 1a). Relatively similar gene structures are observed for closely related genes, although few exceptions are found (Fig. 1b). Concerning their exon–intron organization, although comprising between two (*EgrLAC43*) and eight (*EgrLAC28* and *EgrLAC31*) exons, the majority of the *EgrLAC* genes contains 5–6 exons.

To assess potential duplication events, a locus search in PGDD was performed, and as a result, at least nine *EgrLAC* gene pairs were observed (Table 2). Among them, only *EgrLAC28* and its paralogue (*EgrLAC31*) are located on the same chromosome (chromosome 6). However, since both genes are more than 100 kb apart from each other, the possibility of a tandem duplication event is discarded. To further investigate the mechanisms involved in gene duplication evolution after divergence, the Ka/Ks ratio of each *EgrLAC* gene pairs was calculated (Table 2). The resulting data revealed that all *EgrLAC* gene pairs have Ka/Ks ratios < 1, suggesting that these genes had undergone purifying selection pressure with limited functional divergence after duplication.

The encoded *EgrLAC* proteins (500–600 amino acid residues in length) share extensive sequence identity and predicted molecular masses ranging from 60 to 64 kDa (Table 1). As already mentioned, all the identified sequences, with the exception of *EgrLAC21* and *EgrLAC17*, possess the three types of copper binding sites typically found in known laccase proteins (Fig. S1). The theoretical isoelectric points ranged from 4.81 (*EgrLAC32*) to 9.9 (*EgrLAC47*) (Table 1). The majority of the identified laccases are secreted proteins, but some of them are predicted to be vacuolar or associated with the plasma membrane (Table 1).

Phylogenetic relationships of the identified *Eucalyptus* laccases

To address the evolutionary relationships between the laccases found in *Eucalyptus* and those present in *A. thaliana*

(*AtLAC*) and *P. trichocarpa* (*PtLAC*), a phylogenetic tree was constructed from their corresponding amino acid sequence alignments. Consistent with previous published data (Turlapati et al. 2011; Lu et al. 2013), the *Eucalyptus* laccases were found to be unevenly distributed among six arbitrary groups (Fig. 2). Groups 1 and 4 are the largest ones, comprising 26 and 15 *EgrLACs*, respectively. On the other hand, Group 6 contains only a single *Eucalyptus* laccase (namely *EgrLAC18*).

Eucalyptus laccases with miR397 target sites

Data from a previous study have demonstrated that a subset of 29 laccase genes from the model woody species *P. trichocarpa* are predicted targets of miR397 (Lu et al. 2013), a known negative regulator of lignin biosynthesis (Lu et al. 2013; Wang et al. 2014). Among these *PtLAC* genes, 17 were down-regulated in miR397a-overexpressing transgenic poplar plants showing reduced lignin content (Lu et al. 2013). To identify the *EgrLAC* family members predicted to be targets of miR397, a target screen was performed using psRNATarget. As a result, of the 54 *EgrLAC* genes, 17 were predicted to be potential miR397 family targets (Table 3), indicating that these genes are post-transcriptionally regulated. Among the identified miR397-targeted *EgrLAC* genes, nine of them are predominantly expressed in vascular tissues and most likely implicated in lignification.

EgrLAC expression in different *Eucalyptus* organs/tissues

A publicly available RNA-seq data generated for six different vegetative organs/tissues collected from a *Eucalyptus* hybrid was first used to dissect the tissue-specific expression of the identified *EgrLAC* genes (Fig. 3). According to this dataset, several *EgrLAC* genes exhibited preferential expression in vascular tissues, suggesting a possible involvement with lignification. Among them, *EgrLAC1*, *EgrLAC4*, *EgrLAC10*, *EgrLAC11*, *EgrLAC12* and *EgrLAC20* were particularly enriched in the xylem, while *EgrLAC17* and *EgrLAC6* were preferentially or exclusively expressed in the phloem. Two additional genes (*EgrLAC7* and *EgrLAC39*) were found to be specifically expressed, but at low levels, in shoot tips.

A number of *EgrLAC* genes displayed expression in leaves and, in this case, distinct patterns between juvenile and adult leaf tissues were noted. For example, *EgrLAC24* and *EgrLAC30*, although highly expressed in shoot tips, were also enriched in young leaf tissues, while *EgrLAC41* and *EgrLAC51* were abundantly expressed in adult leaves. On the other hand, *EgrLAC27* and *EgrLAC50* exhibited relatively uniform expression in both young and adult leaves. Additionally, five genes (*EgrLAC28*, *EgrLAC35*, *EgrLAC36*, *EgrLAC44* and *EgrLAC54*) were expressed,

Table 1 General information of the identified *E. grandis* laccase genes and their deduced proteins

Gene	ID	Chromosomal location	Protein				
			Length (aa)	pI	MW (kDa)	Sub-cellular localization	
<i>EgrLAC1</i>	A01282	01	571	9.13	63.06	Secretory	
<i>EgrLAC2</i>	B00869	02	553	9.19	61.67	Secretory	
<i>EgrLAC3</i>	B00870	02	553	6.87	61.35	Secretory	
<i>EgrLAC4</i>	B02316	02	558	9.55	61.39	Secretory	
<i>EgrLAC5</i>	B02428	02	585	8.57	64.68	Secretory	
<i>EgrLAC6</i>	B02750	01	571	8.94	62.88	Secretory	
<i>EgrLAC7</i>	B02751	01	573	6.02	62.33	Secretory	
<i>EgrLAC8</i>	B02752	01	571	8.86	62.70	Secretory	
<i>EgrLAC9</i>	B02753	01	573	9.11	63.28	Secretory	
<i>EgrLAC10</i>	B02794	01	556	9.48	60.79	Non-secretory	
<i>EgrLAC11</i>	B02795	01	556	9.36	60.81	Non-secretory	
<i>EgrLAC12</i>	B02799	01	556	9.39	60.75	Non-secretory	
<i>EgrLAC13</i>	B04003	02	556	9.4	60.76	Secretory	
<i>EgrLAC14</i>	C02745	03	575	8.68	62.93	Secretory	
<i>EgrLAC15</i>	C03118	03	248	7.63	28.32	Secretory	
<i>EgrLAC16</i>	C03119	03	240	5.67	26.36	Secretory	
<i>EgrLAC17</i>	D00739	05	373	8.46	41.42	Secretory	
<i>EgrLAC18</i>	E00040	05	552	6.85	61.28	Secretory	
<i>EgrLAC19</i>	E03932	05	561	5.4	62.35	Secretory	
<i>EgrLAC20</i>	E03935	05	561	4.86	62.4	Secretory	
<i>EgrLAC21</i>	E04215	05	424	6.63	46.97	Secretory	
<i>EgrLAC22</i>	E04216	05	561	6.13	62.3	Secretory	
<i>EgrLAC23</i>	E04217	05	546	5.97	60.58	Secretory	
<i>EgrLAC24</i>	E04218	05	562	4.86	62.5	Secretory	
<i>EgrLAC25</i>	E04219	05	519	5.66	57.65	Secretory	
<i>EgrLAC26</i>	E04220	05	562	4.88	62.3	Secretory	
<i>EgrLAC27</i>	E04221	05	545	4.79	59.93	Secretory	
<i>EgrLAC28</i>	F02641	06	642	6.74	70.85	Secretory	
<i>EgrLAC29</i>	F02650	06	553	6.52	63.12	Secretory	
<i>EgrLAC30</i>	F02652	06	556	8.26	65.19	Secretory	
<i>EgrLAC31</i>	F04154	06	541	6.72	60.12	Secretory	
<i>EgrLAC32</i>	F04160	06	635	4.81	71.45	Secretory	
<i>EgrLAC33</i>	F04163	06	598	5.01	66.88	Secretory	
<i>EgrLAC34</i>	G03028	07	561	8.49	61.54	Non-secretory	
<i>EgrLAC35</i>	G03098	07	570	7.63	62.25	Secretory	
<i>EgrLAC36</i>	H03315	08	562	7.01	61.96	Secretory	
<i>EgrLAC37</i>	J00166	10	570	5.25	62.41	Secretory	
<i>EgrLAC38</i>	J00532	10	585	8.92	64.7	Secretory	
<i>EgrLAC39</i>	J00888	10	487	6.92	54.63	Non-secretory	
<i>EgrLAC40</i>	J00889	10	541	7.63	60.69	Non-secretory	
<i>EgrLAC41</i>	J01857	10	575	6.09	63.41	Secretory	
<i>EgrLAC42</i>	J01941	10	524	8.87	57.49	Non-secretory	
<i>EgrLAC43</i>	J03171	10	266	7.65	29.21	Secretory	
<i>EgrLAC44</i>	K02996	11	562	9.26	61.55	Non-secretory	
<i>EgrLAC45</i>	K03008	11	573	8.99	62.76	Secretory	
<i>EgrLAC46</i>	K03009	11	548	9	60.44	Secretory	
<i>EgrLAC47</i>	K03111	11	589	9.9	64.96	Secretory	
<i>EgrLAC48</i>	L00355	scaffold_16:11569f	545	8.65	60.15	Secretory	
<i>EgrLAC49</i>	L00359	scaffold_16:43401f	585	8.78	64.72	Secretory	

Table 1 (continued)

Gene	ID	Chromosomal location	Protein			
			Length (aa)	pI	MW (kDa)	Sub-cellular localization
<i>EgrLAC50</i>	L01343	scaffold_173:3472r	560	4.91	61.88	Secretory
<i>EgrLAC51</i>	L01345	scaffold_173:2436f	527	6.11	58.68	Secretory
<i>EgrLAC52</i>	L01441	scaffold_209:43469f	563	6.8	63	Non-secretory
<i>EgrLAC53</i>	L01734	scaffold_327:30946r	556	9.48	60.8	Non-secretory
<i>EgrLAC54</i>	L02310	scaffold_774:703f	445	5.5	49.24	Secretory

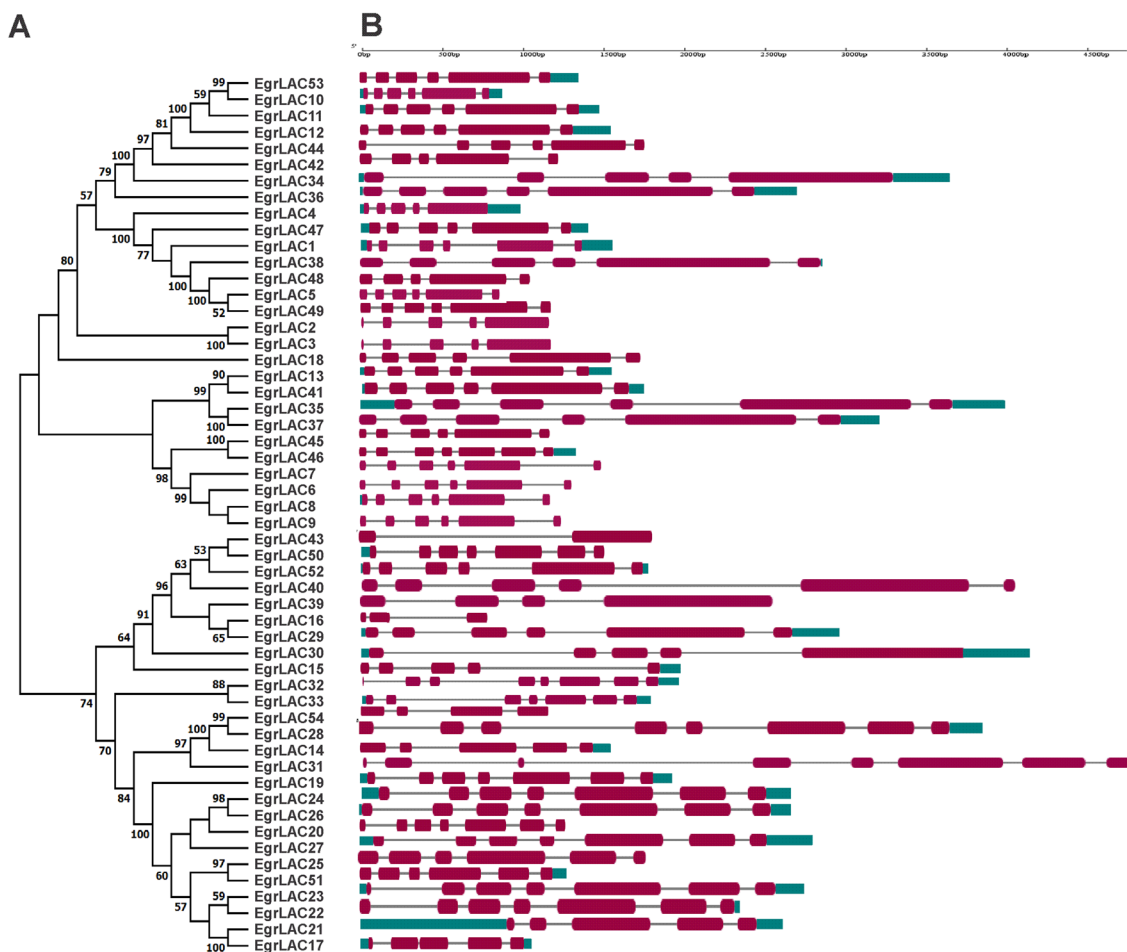


Fig. 1 Phylogenetic relationships and schematic representation of the exon/intron organization of the identified *EgrLAC* genes. **a** Neighbor-joining tree (1000 replicas) constructed using the full-length amino acid sequences of the 54 *EgrLAC*s. Only bootstrap values above 50% are shown. **b** Exon/intron structure of the corresponding genes. The

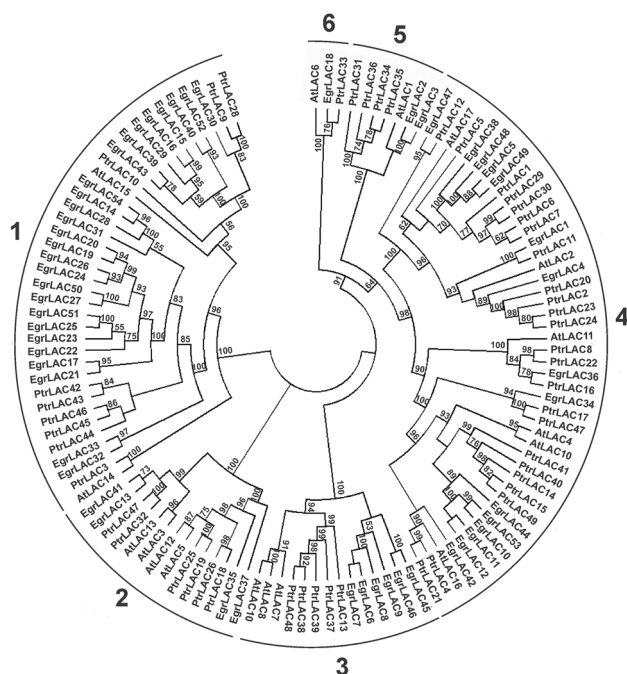
red rectangles represent exons and thin black lines represent introns. The untranslated regions are depicted in blue. Lengths in nucleotides are shown in the upper part. The graphic representation was displayed using GSDS v2.0

albeit at different levels, in all profiled organs/tissues. In contrast, 11 genes (*EgrLAC2*, *EgrLAC5*, *EgrLAC8*, *EgrLAC9*, *EgrLAC15*, *EgrLAC16*, *EgrLAC31*, *EgrLAC43*, *EgrLAC48*, *EgrLAC49* and *EgrLAC52*) showed no detectable expression, indicating that they might be expressed in organs/tissues other than those sampled in Phytozome or may not be functional. In fact, when looking to a more

comprehensive organ/tissue expression atlas from *Eucalyptus* (EucGenIE database), the expression of *EgrLAC31* was found to be restricted to flower buds at stage 1 (Vining et al. 2014), while *EgrLAC2*, *EgrLAC5*, *EgrLAC8*, *EgrLAC9*, *EgrLAC15*, *EgrLAC16*, *EgrLAC43* and *EgrLAC48* were found to be preferentially expressed in roots. Still based on the EucGenIE database, both *EgrLAC2* and

Table 2 Synonymous (Ks) and non-synonymous (Ka) substitution rates of the predicted *EgrLAC* gene pairs

Pair	Gene 1	Gene 2	Ka	Ks	Ka/Ks
1	<i>EgrLAC1</i>	<i>EgrLAC4</i>	0.26	1.75	0.1486
2	<i>EgrLAC4</i>	<i>EgrLAC47</i>	0.25	1.52	0.1645
3	<i>EgrLAC6</i>	<i>EgrLAC45</i>	0.20	1.32	0.1515
4	<i>EgrLAC10</i>	<i>EgrLAC42</i>	0.19	1.49	0.6711
5	<i>EgrLAC10</i>	<i>EgrLAC44</i>	0.13	1.11	0.1171
6	<i>EgrLAC28</i>	<i>EgrLAC31</i>	0.18	4.24	0.0424
7	<i>EgrLAC34</i>	<i>EgrLAC42</i>	0.24	1.56	0.1538
8	<i>EgrLAC34</i>	<i>EgrLAC44</i>	0.22	1.26	0.1746
9	<i>EgrLAC35</i>	<i>EgrLAC37</i>	0.14	1.94	0.0721

**Fig. 2** Phylogenetic relationships (maximum likelihood) between laccases from *Eucalyptus grandis* (Egr), *Arabidopsis thaliana* (At), and *Populus trichocarpa* (Ptr). The branch support was obtained by the aLRT method, and nodes supported by aLRT < 50% were hidden. The different laccase groups (1–6) are indicated

EgrLAC16 were as well expressed in flowers (stage 1 and stage 2, respectively).

To obtain a more complete picture of the spatial expression of the *EgrLAC* genes, a complementary analysis was performed using an in-house unpublished RNA-seq data obtained from three vegetative organs/tissues (leaf, stem and root) collected from *E. grandis* plantlets (Fig. 4). Confirming our expectations, several *EgrLAC* genes displayed enriched expression in roots, and nine among them (*EgrLAC8*, *EgrLAC9*, *EgrLAC13*, *EgrLAC23*, *EgrLAC29*, *EgrLAC41*, *EgrLAC42*, *EgrLAC45* and *EgrLAC52*) were

highly root-specific. Consistent with our data, 16 of the 17 *AtLAC* genes identified in *Arabidopsis* were also abundantly expressed in the root, which is the organ with the highest level of laccase expression (Cai et al. 2006). In contrast, the most highly expressed genes in leaves were *EgrLAC24*, *EgrLAC30* and *EgrLAC32*, while *EgrLAC53* was prevalent in the stem. A subset of genes (*EgrLAC1*, *EgrLAC4*, *EgrLAC10*, *EgrLAC11*, *EgrLAC12*, *EgrLAC34*, *EgrLAC35*, *EgrLAC44*, *EgrLAC47* and *EgrLAC54*) showed enriched expression in both stem and root, whereas another subset (*EgrLAC27*, *EgrLAC28*, *EgrLAC32* and *EgrLAC33*) was expressed at different levels in all tested organs (Fig. 4).

Interestingly, the majority of the putative *EgrLAC* gene pairs, with the exceptions of *EgrLAC34/EgrLAC44*, *EgrLAC10/EgrLAC44* and *EgrLAC35/EgrLAC37* exhibited divergent expression patterns, thus suggesting functional diversification.

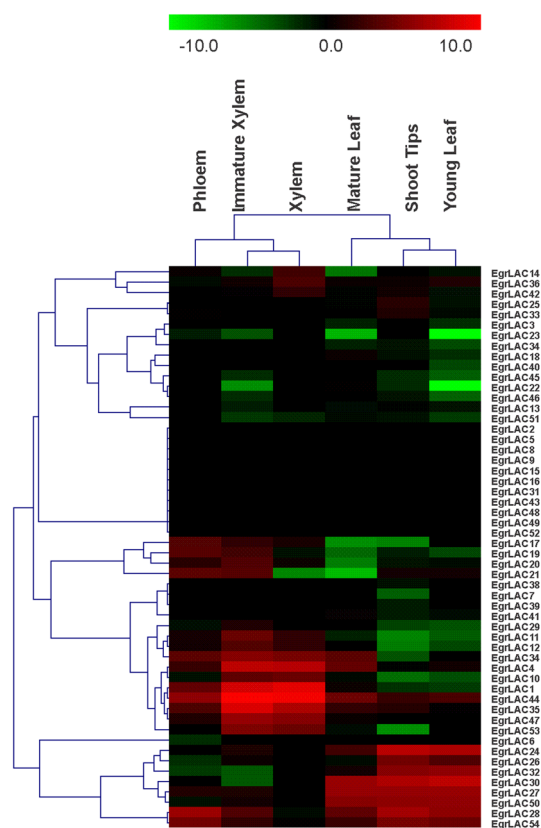
To validate the transcriptional profiles shown by the RNA-Seq data, seven randomly selected *EgrLAC* genes (*EgrLAC2*, *EgrLAC4*, *EgrLAC30*, *EgrLAC32*, *EgrLAC38*, *EgrLAC40* and *EgrLAC52*) were analyzed by RT-qPCR in three *E. grandis* organ/tissue samples (root, stem and leaf). These genes were chosen based on their distinctive organ/tissue expression patterns revealed by RNA-Seq (*EgrLAC2*, *EgrLAC4* and *EgrLAC52* in roots; *EgrLAC30* and *EgrLAC32* in leaves; *EgrLAC38* and *EgrLAC40*, which are ubiquitously expressed at low levels), to have at least two representative genes per sampled organ/tissue. Although some divergences could be observed, most of the expression patterns determined by the RNA-Seq analysis were validated via RT-qPCR (Fig. S2). The obtained results confirmed, for example, the preferential expression of *EgrLAC2* in roots, in spite of a relatively low expression in both leaf and stem. Likewise, the root-predominant expression of *EgrLAC4* was validated, but residual expression in both leaf and stem was also detected (Fig. 3), which is in accordance with the RNA-seq data from Phytozome. Moreover, the preferential expression of both *EgrLAC30* and *EgrLAC32* in leaf tissues as revealed by all inspected RNA-Seq data was corroborated by the RT-qPCR validation. Also consistent with our in-house RNA-seq profiling, the RT-qPCR analysis revealed an apparent root-specific expression of *EgrLAC52*. Intriguingly, the expression of both *EgrLAC38* and *EgrLAC40* was undetectable by RT-qPCR. These genes, however, were discreetly expressed in different organs/tissues as revealed by the RNA-seq profiling (Figs. 3, 4).

***EgrLAC* expression in response to oxidative and osmotic stresses**

Since laccases have been proposed to play a role in the acquisition of abiotic stress tolerance in plants (Cai et al. 2006; Cho et al. 2014; Ma et al. 2015; Liu et al. 2017), we

Table 3 List of the *EgrLAC* genes with putative miR397 target sites

Gene	Predicted miR397 family targets	miRNA length	Type of inhibition
<i>EgrLAC1</i>	miR397, miR397a, miR397b, miR397a-5p, miR397b-5p	21–22	Cleavage
<i>EgrLAC2</i>	miR397*, miR397a, miR397b, miR397b-5p*	20–21	Cleavage translation
<i>EgrLAC3</i>	miR397*, miR397a, miR397b, miR397a-5p, miR397b-5p*	20–22	Cleavage translation
<i>EgrLAC4</i>	miR397, miR397a, miR397b, miR397a-5p, miR397b-5p	20–22	Cleavage
<i>EgrLAC6</i>	miR397, miR397a, miR397b, miR397a-5p, miR397b-5p	20–22	Cleavage
<i>EgrLAC7</i>	miR397, miR397b	21	Cleavage
<i>EgrLAC8</i>	miR397, miR397a, miR397b, miR397b-5p	20–21	Cleavage
<i>EgrLAC10</i>	miR397a-5p, miR397b-5p	21	Cleavage
<i>EgrLAC11</i>	miR397, miR397a, miR397b, miR397a-5p, miR397b-5p	21–22	Cleavage
<i>EgrLAC12</i>	miR397, miR397a, miR397b, miR397a-5p, miR397b-5p	21–22	Cleavage
<i>EgrLAC13</i>	miR397, miR397a, miR397b, miR397b-5p	20–21	Cleavage
<i>EgrLAC35</i>	miR397, miR397a, miR397b, miR397b-5p	20–21	Cleavage
<i>EgrLAC42</i>	miR397a-5p, miR397b-5p	21	Cleavage
<i>EgrLAC44</i>	miR397, miR397a, miR397a-5p, miR397b-5p	21–22	Cleavage
<i>EgrLAC45</i>	miR397, miR397a, miR397a-5p, miR397b-5p	20–21	Cleavage
<i>EgrLAC46</i>	miR397, miR397a, miR397b, miR397b-5p	20–21	Cleavage
<i>EgrLAC53</i>	miR397a-5p, miR397b-5p	21	Cleavage

**Fig. 3** Heatmap representation of the *EgrLAC* expression profiles in different *E. grandis* organs/tissues. The heatmap was generated employing normalized FPKM values available at Phytozome. After the addition of a pseudo-count of 1, all FPKM values were $\log_2$ -transformed [$\log_2(\text{FPKM} + 1)$]. Genes were clustered according to their expression profiles

decided to investigate whether the identified *EgrLAC* genes might be responsive to abiotic stresses. For this, the expression profiles of five selected *EgrLAC* genes (*EgrLAC4*, *EgrLAC14*, *EgrLAC35*, *EgrLAC45* and *EgrLAC52*) in response to oxidative (H_2O_2 -induced) and osmotic (PEG-induced) stress, respectively, were determined using RT-qPCR. It should be emphasized that these genes, which show preferential expression in stems and roots, were selected based on their potential responsiveness to different stresses according to previous in silico analyses performed in the private *Scylla* EST database (not shown).

Despite showing differences in the magnitude and timing of induction, all tested *EgrLAC* genes exhibited increased expression in response to PEG-imposed osmotic stress in the stem (Fig. 5a). The only obvious exception was the root-specific *EgrLAC52* gene. On the other hand, solely *EgrLAC14*, *EgrLAC35* and *EgrLAC45* were significantly induced by osmotic stress in the roots (Fig. 5b). The expression of the investigated genes, with the exception of *EgrLAC52*, was also induced in the stem of plantlets submitted to oxidative stress (Fig. 6a). In this case, peak expression levels were mainly observed after 12 h (*EgrLAC4*) and 24 h (*EgrLAC14*, *EgrLAC35* and *EgrLAC45*) of stress imposition. In contrast, only *EgrLAC14*, *EgrLAC45* and *EgrLAC52* were significantly up-regulated by oxidative stress in the roots, with *EgrLAC45* figuring as the most responsive (Fig. 6b). Among the investigated genes, *EgrLAC45* and *EgrLAC14* figured as the only genes significantly induced in both organs and under both stress conditions. These results suggest that, as already reported in other species, the investigated laccases might play a role in *Eucalyptus* adaptation to abiotic stresses.

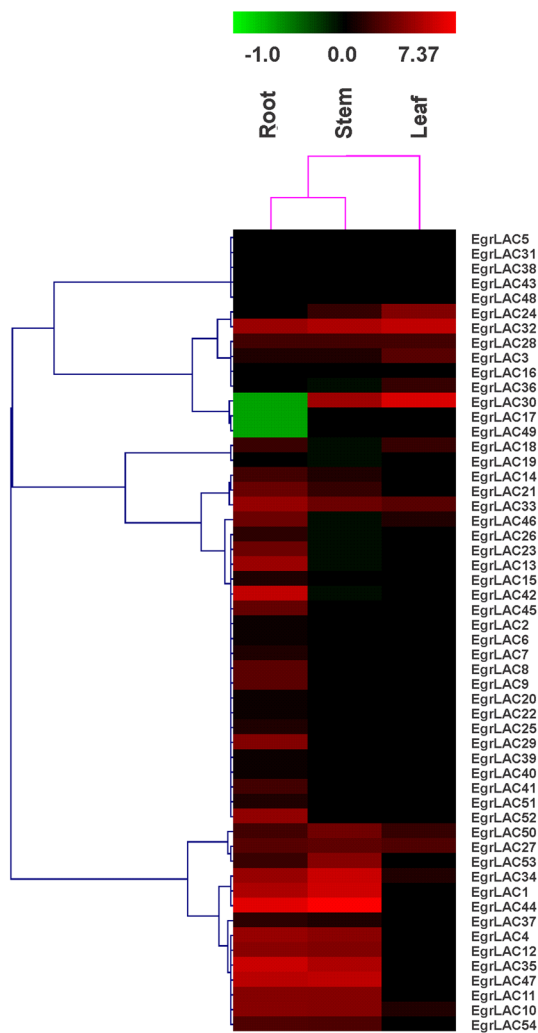


Fig. 4 Heatmap representation of the *EgrLAC* expression profiles in root, stem and leaf samples collected from 4-month-old *E. grandis* plantlets. The heatmap was generated employing normalized log₂-transformed RPKM values added of a pseudo-count of 1 [$\log_2(\text{RPKM}+1)$]. Genes were clustered according to their expression profiles

GUS expression driven by a selected *EgrLAC* gene promoter

To extend and further validate the observations made using the aforementioned expression data, we selected an abiotic stress-responsive *EgrLAC* gene (namely *EgrLAC4*) showing high sequence identity (70% at the amino acid level) with *AtLAC17* and *AtLAC2*, known to be functionally involved in lignin biosynthesis and in the Arabidopsis response to dehydration stress, respectively, and investigated the expression pattern driven by its promoter in Arabidopsis. For that, *Arabidopsis* transgenic lines harboring an *EgrLAC4 promoter::GUS* reporter cassette were generated.

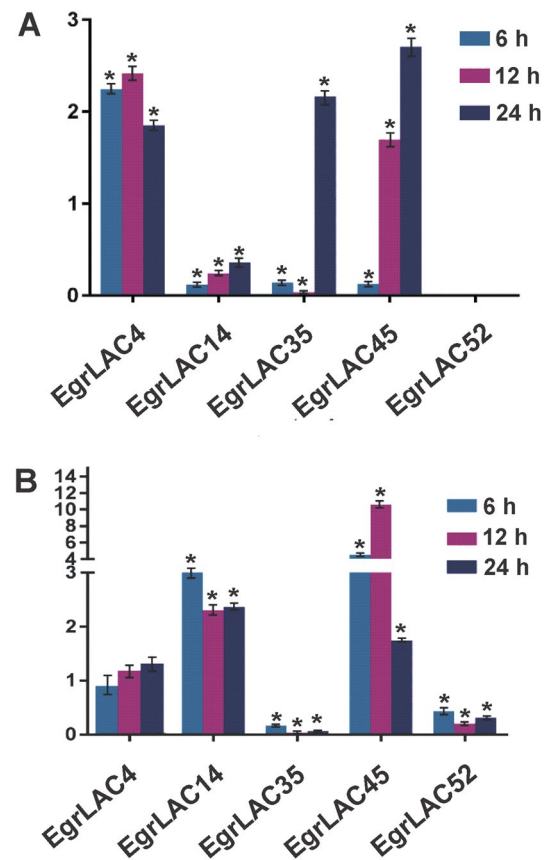


Fig. 5 Expression patterns of selected *EgrLAC* genes under PEG-induced osmotic stress. **a** Stem, **b** root. *EgrLAC* expression at each time point after PEG treatment was calculated relative to that of the untreated control at the same time point, and normalized to the expression of *EgTEF*. Average data with standard errors from three replicates is presented (* $p < 0.05$)

Histochemical analysis of intact 7-day-old transgenic seedlings ($n = 3$) revealed *EgrLAC4* promoter activity in the vascular tissues of the cotyledons and hypocotyls (Fig. 7a). In mature leaves, the GUS signal was detected along the veins and in trichomes (Fig. 7b, c). In addition, GUS staining was also visualized in vascular tissues along the root (not shown). The observed expression contrasted with the ubiquitous pattern of GUS expression observed in Arabidopsis seedlings transformed with a *p35S::GUS* cassette (Fig. 7f). As expected, no GUS expression was detected in WT seedlings (Fig. 7g). Cross-sections of stained roots and leaves revealed that GUS activity was mostly associated with the phloem (Fig. 7d, e), thus confirming main expression in the vasculature. Strikingly, GUS staining was also occasionally detected in the leaf cuticle (Fig. 7e). Altogether, these results indicate that the amplified *EgrLAC4* upstream fragment (1.38 kb) was sufficient to confer vascular expression in Arabidopsis, which is consistent with the endogenous gene expression profile observed in *Eucalyptus* (Fig. 3).

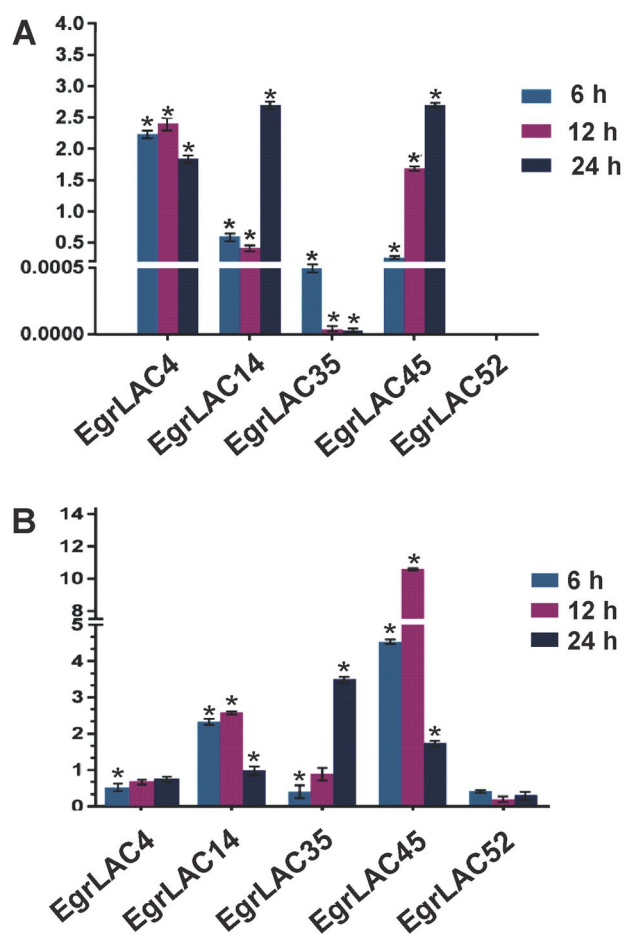


Fig. 6 Expression patterns of selected *EgrLAC* genes under H_2O_2 -induced oxidative stress. **a** Stem, **b** root. *EgrLAC* expression at each time point after H_2O_2 treatment was calculated relative to that of the untreated control at the same time point, and normalized to the expression of *EgTEF*. Average data with standard errors from three replicates is presented (* $p < 0.05$)

Discussion

Due to their roles in lignin biosynthesis and secondary cell wall formation, plant laccases are considered potential targets for different biotechnological applications, especially in woody species (Wang et al. 2015a). Interestingly, recent studies have also demonstrated that the function of plant laccases extends beyond lignification to a variety of physiological processes, including biotic and abiotic stress tolerance (Cho et al. 2014; Ma et al. 2015; Liu et al. 2017; Hu et al. 2018). Nevertheless, since in plants the laccase family is quite extensive, the functional role played by the individual isoforms remains to be elucidated. In this study, with the aim to improve our knowledge about laccases in woody species, we established a complete picture of the laccase gene family in *Eucalyptus*, an economically important and fast-growing forest tree.

Although as many as 88 sequences were previously postulated to encode laccases in the genome of *E. grandis* (Myburg et al. 2014), indicating an important expansion of this family compared to other species, our survey restricted the size of this family to 54 members. When compared to other plant species, the number of putative genes found in *Eucalyptus* is close to the total identified in *P. trichocarpa* (49 genes; Lu et al. 2013), *L. usitatissimum* (45 genes; Le Roy et al. 2017) and in two diploid *Gossypium* spp. (44 and 46 genes in *G. arboreum* and *G. raimondii*, respectively; Balasubramanian et al. 2016), but differs considerably from the number present in *A. thaliana* (17 genes) and in monocotyledonous species such as *Sorghum bicolor* (27 genes; Wang et al. 2017), *Saccharum officinarum* (12; Cesarino et al. 2013) and *Brachypodium distachyon* (29 genes; Wang et al. 2015b). To date, however, the cultivated allotetraploid cotton *G. hirsutum* is the species with the highest number of LAC genes in its genome (84 genes in total; Balasubramanian et al. 2016).

Chromosomal localization analysis revealed that the identified laccase genes were distributed in 9 out of 11 chromosomes of *Eucalyptus*. Regarding their genomic organization, the annotated *EgrLAC* genes showed a conserved exon–intron structure, with an exon number ranging from 2 to 8. This organization is equally found in LAC genes from other plant species (Cai et al. 2006; Liu et al. 2017; Wang et al. 2017). Interestingly, the multicopper oxidase type 1 motif is encoded by exon 5, which is the longest and most conserved exon.

According to our phylogenetic analysis, all six laccases groups previously described in *A. thaliana* (McCaig et al. 2005; Cai et al. 2006) are also observed in *Eucalyptus*. It is generally accepted that the clustering of a gene in a given group is indicative of redundant roles and might be used to predict potential laccase function (McCaig et al. 2005). In this context, 15 *EgrLAC*s were clustered in the so-called Group 4, which comprises laccases reported to be involved in lignin biosynthesis, especially during vascular development, such as AtLAC17, AtLAC4, AtLAC11 and PtLAC2 (Ranocha et al. 2002; Berthet et al. 2011; Zhao et al. 2013; Bryan et al. 2016). On the other hand, Groups 1 and 3 comprise, respectively, Arabidopsis laccases reported to be implicated in the oxidation of flavonoids such as AtLAC15 (Cai et al. 2006; Liang et al. 2006a) and those responsive to environmental cues such as AtLAC7, AtLAC8 and AtLAC9 (McCaig et al. 2005).

Noteworthy, 11 out of 15 *EgrLAC* genes included in Group 4 are preferentially associated with vascular tissues, and more particularly, with the secondary xylem where secondary cell wall formation and lignin biosynthesis take place (Plomion et al. 2001). This spatial expression pattern is consistent with a putative role for these genes in lignification during wood formation and in line with the reported

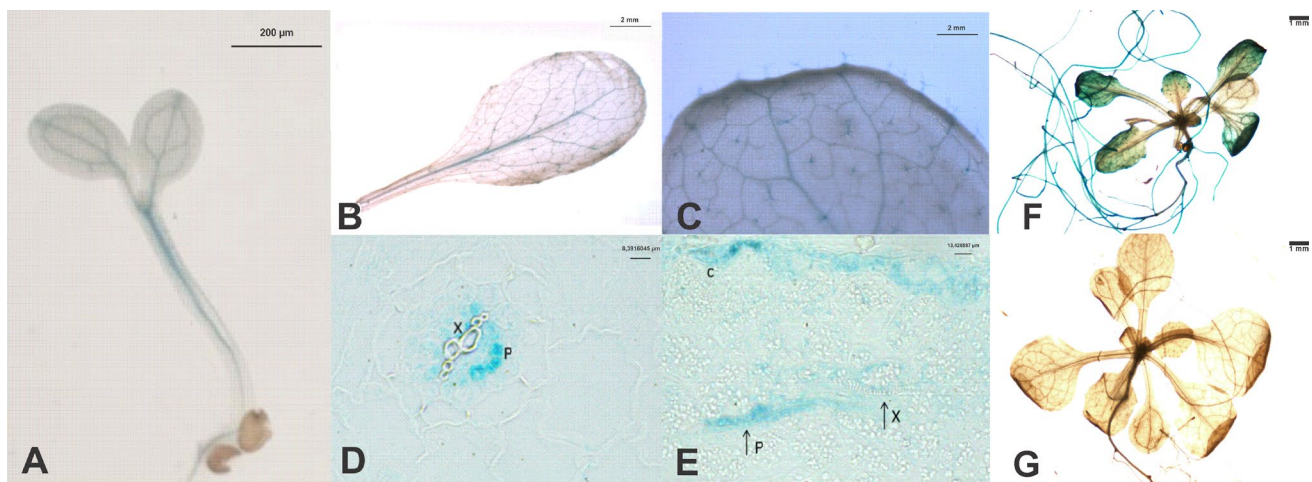


Fig. 7 Histochemical analysis of GUS activity in different organs/tissues of *Arabidopsis* (Col-0) transformed with the *EgrLAC4 promoter::GUS* expression cassette. **a** 7-day-old seedling; **b** 20-day-old leaf; **c** trichomes; **d** cross-section of a stained root; **e** cross-section

of a stained leaf. *P* phloem; *X* xylem; *C* cuticle; **f** positive control (*p35S::GUS*, 15 day-old seedling); **g** negative control (wild-type Col-0, 15 day-old seedling). Images of a representative transgenic *Arabidopsis* line are shown

prominent expression of laccase genes in developing xylem of *Eucalyptus* (Hefer et al. 2015). However, caution should be taken while interpreting such RNA-Seq based expression results since the interrogated tissue and organ-specific expression datasets were generated from samples collected from a single eucalypt tree genotype (Mizrachi et al. 2010). Equally interesting, among the *EgrLAC* genes predicted to be targeted by miR397, which negatively regulates the biosynthesis of lignin by modulating laccase expression (Lu et al. 2013; Wang et al. 2014), eight of them were present in the aforementioned Group 4.

An additional evidence for a possible involvement of Group 4 members in lignin biosynthesis came from the histochemical analysis of the promoter activity of *EgrLAC4*, a Group 4 gene and a close homologue of the *Arabidopsis* lignin-associated and vascular-expressed gene *AtLAC17*. In this context, although predicted from public RNA-Seq data to be preferentially expressed in the xylem, *EgrLAC4* promoter-driven GUS expression was observed mainly in the vascular bundle, and most preferentially in the phloem. A similar enriched expression in the phloem of inflorescence stem tissues was also observed for the stress-inducible *AtLAC8* gene belonging to Group 3 (Turlapati et al. 2011). The phloem expression of *EgrLAC4* suggests a specialized role during the lignification of the phloem cells. In support to this, using *Populus* as a model, Jin et al. (2018) observed that the relative concentration and associated distribution of lignin monomers fluctuate according to the phloem cell location, a feature that would require distinct tissue-specific laccases.

Apart from the mentioned phloem- or xylem-enriched expression, the identified *EgrLAC* genes also displayed

preference to a particular organ. In this context, as already reported in other plant species (Cai et al. 2006; Liu et al. 2017), approximately 58% of the *EgrLAC* family members showed predominant expression in root, while about 35% were preferentially expressed in the stem. Additional experimentation will be necessary to establish whether these genes act redundantly in these organs or might serve a different function. In this regard, one can speculate that these *EgrLACs*, although possibly possessing overlapping substrate specificities, might be functional in different cell types and under different physiological conditions. On the other hand, given that both root and stem contain a significant amount of lignified tissues (Barros et al. 2015), a major role for these *EgrLACs* in lignification is still envisioned.

Another point worthy of note is the detected stress-induced expression of a set of *EgrLAC* members, whose encoded products were distributed in different phylogenetic groups. These results are in agreement with previous reports (Liang et al. 2006b; Liu et al. 2017; Hu et al. 2018) and reinforce the notion that plant laccases are involved in different physiological processes. Interestingly, the examined *EgrLAC* genes have been shown to be responsive to osmotic and oxidative stress, which are postulated to affect lignin content and composition (Moura et al. 2010; Rui et al. 2016). Likewise, drought stress has been reported to alter lignin amount and composition in *Eucalyptus* (Moura-Sobczak et al. 2011). Regarding the acquisition of biotic and abiotic stress tolerance by the modulation of plant laccase genes, this has been correlated with the production of phenolic polymers and enhanced flavonoid oxidation (Cho et al. 2014), and the accumulation of jasmonic acid and secondary metabolites derived from the phenylpropanoid pathway (Hu et al.

2018). Overall, our data provide evidence for a contribution of specific EgrLAC isoforms in the *Eucalyptus* adaptation to stressful conditions.

Conclusions

In all, our results provide a more complete knowledge of the *Eucalyptus* laccase gene family. Considering that lignin is the most important factor limiting the conversion of plant biomass to pulp or biofuels, fine tuning its content thorough the modulation of laccase genes in plants represents a promising strategy. The fact that an impact on plant tolerance to biotic and abiotic stresses could be also attained by the genetic manipulation of such genes is equally interesting. In this context, although future studies will be necessary to sort out the exact role played by each of the 54 EgrLAC family members, our data provide *Eucalyptus* biotechnology with interesting candidate genes for the implementation of such approaches.

Author contribution statement Conceived and designed the experiments: MLCA, LCF, CLM and IGM; performed the experiments: MLCA, LCF, IRW and MCPF-F; analyzed the data: IRW, GTV and IGM; wrote the paper: MLCA and IGM.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Data archiving statement Sequence Read Archive accessions for RNA-seq data are: SRR1481484; SRR1481485; SRR1481487; SRR1481488; SRR1481490 and SRR1481491.

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